This section will cover how you can manage external access to applications running in a Kubernetes cluster using the Ingress API.

NOTE This section assumes you have gone through the tasks listed in the previous "Polar Labs" sidebar and prepared Edge Service for deployment on Kubernetes.

9.5.1 Understanding Ingress API and Ingress Controller

When it comes to exposing applications inside a Kubernetes cluster, we can use a Service object of type `ClusterIP`. That's what we've done so far to make it possible for Pods to interact with each other within the cluster. For example, that's how Catalog Service Pods can communicate with the PostgreSQL Pod.

A Service object can also be of type `LoadBalancer`, which relies on an external load balancer provisioned by a cloud provider to expose an application to the internet. We could define a `LoadBalancer` Service for Edge Service instead of the `ClusterIP` one. When running the system in a public cloud, the vendor would provision a load balancer, assign a public IP address, and all the traffic coming from that load balancer would be directed to the Edge Service Pods. It's a flexible approach that lets you expose a service directly to the internet, and it works with different types of traffic.

The `LoadBalancer` Service approach involves assigning a different IP address to each service we decide to expose to the internet. Since services are directly exposed, we don't have the chance to apply any further network configuration, such as TLS termination. We could configure HTTPS in Edge Service, route all traffic directed to the cluster through the gateway (even platform services that don't belong to Polar Bookshop), and apply further